Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2024-I-Q03] | Part A (Q1-Q5)

Answer: 809

1. The losing positions for the player to move are exactly $n \equiv 0, 2 \pmod 5$. From either residue every legal move reaches $1, 3$, or 4; from $1, 3, 4$ one can respectively remove $1, 1, 4$ to reach a losing residue.

2. The base cases $0, 1, 2, 3, 4$ verify the claim, so induction applies. The required count is $\lfloor 2024/5 \rfloor + \lfloor (2024 - 2)/5 \rfloor + 1 = 404 + 405 = 809$.

Source / 来源

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2010-I-Q12] | Part C (Q11-Q15)

Answer: 243

1. For $m = 243$, suppose there is no monochromatic product, and color 3 red. Then 9 is blue, 81 red, and 243 blue. If 27 is red, $3 \cdot 27 = 81$ fails; if blue, $9 \cdot 27 = 243$ fails. This proves the upper bound.

2. For $m = 242$, color $[3, 8] \cup [81, 242]$ red and $[9, 80]$ blue. Products of two small red numbers lie in $[9, 64]$, hence blue; any product involving a large red number exceeds 242. Products of two blue numbers exceed 80, hence are not blue within the set.

3. This product-free coloring also restricts to every smaller initial set. Therefore the least value is 243.

Source / 来源

Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2012-I-Q11] | Part C (Q11-Q15)

Answer: 373

1. Every move keeps $x+y$ a multiple of three and $x-y$ a multiple of five. Write $x+y=3j, x-y=5k$; integer coordinates require j,k of the same parity. The diamond bound is $\max(|3j|,|5k|)\le 100$, so $|j|\le 33, |k|\le 20$.

2. These constraints are sufficient. Repeating a suitable move reaches some point with each desired integer k. Two positive moves together shift by $(9,9)$; two negative moves shift by $(-15,-15)$. Combining them in counts $(2,1)$ or $(3,2)$ gives diagonal shifts $(3,3)$ or $(-3,-3)$.

3. These shift j by ± 2 without changing k, reaching every required same-parity value. Count even pairs $33\cdot 21$ and odd pairs $34\cdot 20$, for $M=1373$.

4. The remainder is 373.

Source / 来源